ANTARA THAPLIYAL

Hamilton, NJ | 640-225-3875 | thapliyalantara31@gmail.com | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Master's student in Computer Science at Rutgers University with 2 years of production software engineering experience. Skilled in **Python, LangChain, LangGraph, MCP, prompt engineering, and GenAI evaluation**, with hands-on experience building AI Agents, agentic workflows, and RAG-based applications integrating vector stores, APIs, and LLM frameworks. Currently developing a multi-agent AI system with real-time tool connectivity and evaluation pipelines. Seeking to apply generative AI engineering and scalable evaluation skills in a fast-paced, innovation-driven environment.

TECHNICAL SKILLS

Languages & Databases: Python, SQL/MySQL, HTML, CSS
GenAI & Agent Frameworks: LangChain, LangGraph, Langfuse, MCP (Model Context Protocol), OpenAI API, CrewAI, FAISS, Chroma, RAG Architectures
ML Frameworks: PyTorch, TensorFlow, Pandas, NumPy
Core Skills: AI Agent Development, Prompt Engineering & Robustness Testing, Agentic Workflow Design, GenAI Evaluation & Benchmarking, AI Guardrails & Safety, Explainability & Interpretability, Vector Stores/RAG, REST API Integration, Data Pipeline Design, Technical Documentation
Tools & Platforms: Git/GitHub, Jupyter Notebook, Streamlit, VS Code, PyCharm, Microsoft Office
Methodologies: Agile/Scrum, SDLC, Iterative Experimentation & Design

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS), Pune, India *Jul 2021 – Jun 2023*
Software Engineer – Insurance Domain

- Partnered with cross-functional teams to design and deliver new product features for an enterprise platform, driving a **29% lift in sales**.
- Engineered automated batch-processing data pipelines and letter-generation utilities, cutting manual effort by **30%** and accelerating offline processing workflows.
- Built the "Autopay Re-collection" system, integrating database queries and REST API calls for automated loan repayment processing, improving collection efficiency.
- Maintained clear technical documentation, Git-based versioning workflows, and sprint documentation to ensure reproducibility and clarity across engineering teams.

PROJECTS

Shadow Board – Multi-Agent AI Decision System | *Python, CrewAI, MCP, REST APIs, FastAPI, SSE* *2026*

- Architecting a multi-agent AI system with specialized AI Agents (CFO, CMO, Legal, Devil's Advocate) that independently research via API-based tool integrations and engage in structured real-time agentic workflows with evaluation checkpoints.
- Engineered prompt engineering strategies and prompt robustness testing for distinct agent personas, optimizing response quality, safety guardrails, and inter-agent collaboration using CrewAI.
- Designed evaluation harnesses and benchmarking pipelines to assess agent decision quality across diverse scenarios, iterating on agent behaviors through systematic experimentation and documentation.

Capital Finder Bot – RAG-Based AI Chatbot | *LangChain, Langfuse, OpenAI API, OpenAI text embedding, FAISS, Streamlit, Python* *2025*

- Developed a RAG-powered chatbot integrating LangChain, OpenAI GPT-3.5 API, and FAISS vector store for context-aware retrieval, applied prompt engineering and token optimization.
- Developed a RAG pipeline integrating LangChain, OpenAI text-embedding-3-small, and FAISS vector store for context-aware retrieval; implemented RAGAS evaluation measuring faithfulness and answer relevancy, and integrated LangFuse for end-to-end tracing of latency, token usage, and cost per query.

Abilify Clinical Q&A Agent – Agentic RAG System | *Python, LangGraph, LangChain, LangFuse, RAGAS, OpenAI, Chroma* *2026*

- Built a four-agent clinical Q&A system on Abilify's FDA label using LangGraph for orchestration, with specialist agents (Clinical, Drug Interaction, Safety) each using LangChain AgentExecutor with RAG retrieval tools combining parent-child chunking, hybrid BM25 and vector search, and Cohere reranking for precise context retrieval.
- Implemented an inline evaluation agent that autonomously checks faithfulness, completeness, source citation, and medical disclaimer using LLM-as-judge tools, routing back to specialist agents for retry when quality is insufficient; integrated LangFuse for full observability and RAGAS for offline quality monitoring.

Incentive-Compatible Reward Design for Adaptive AI Agents | *Python, Q-Learning, NumPy* *2025*

- Conducted evaluation and benchmarking experiments across 4 reward schemes, training agents under non-stationary conditions with concept drift, demonstrating adaptive recovery and presenting comparative findings on calibration accuracy.

EDUCATION

Rutgers University, New Brunswick, NJ *Expected Dec 2026*
Master of Science in Computer Science | GPA: 4.0

G.B. Pant University of Agriculture & Technology, India *Jun 2021*
Bachelor of Technology in Information Technology

LEADERSHIP & ACTIVITIES

- **Certification:** Supervised Machine Learning: Regression and Classification – Stanford Online & DeepLearning.AI (Coursera), Introduction to Model Context Protocol by Anthropic.
- **Part-time Lecturer:** Rutgers University
- **Astitva NGO (2017–2021):** Dedicated three days weekly to teaching underprivileged children, enhancing educational access, and community wellbeing.